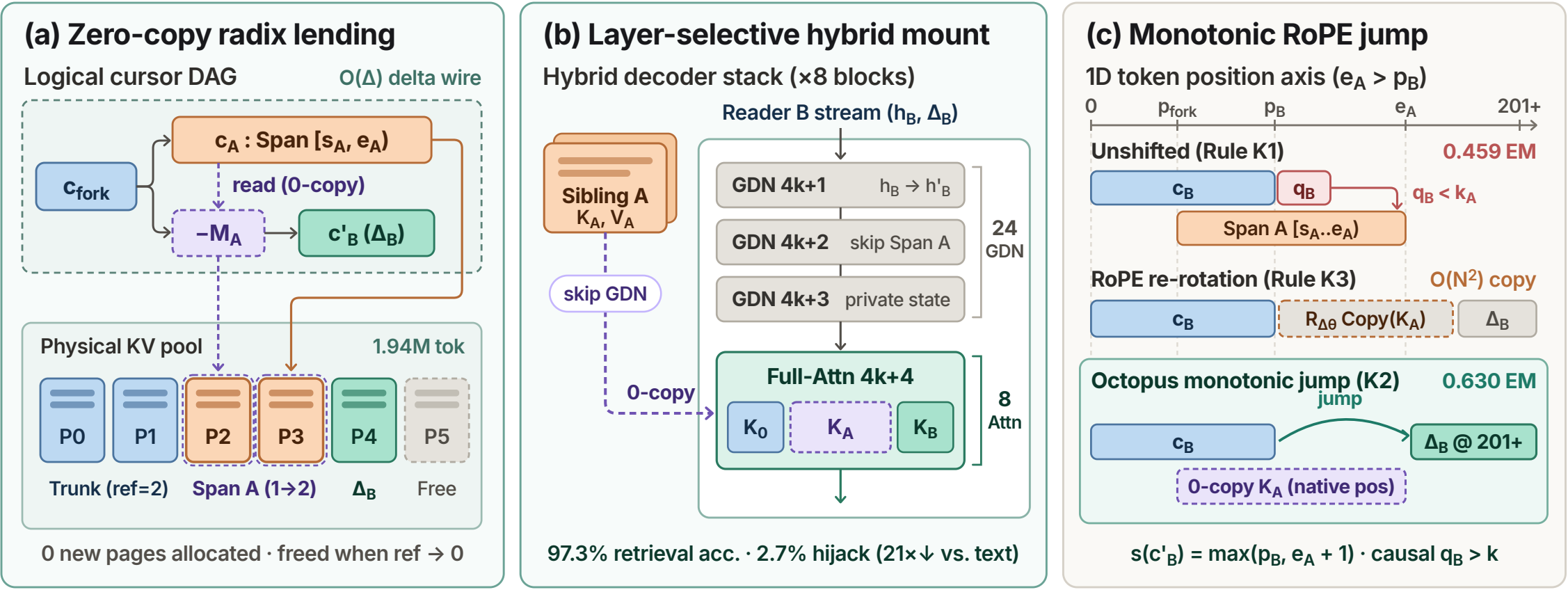
(a) Zero-copy radix lending
Logical cursor DAG
O(Δ) delta wire
c_A : Span [s_A, e_A)
c_fork
read (0-copy)
−M_A
c'_B (Δ_B)
Physical KV pool
1.94M tok
P0
P1
P2
P3
P4
P5
Trunk (ref=2)
Span A (1→2)
Δ_B
Free
0 new pages allocated · freed when ref → 0
(b) Layer-selective hybrid mount
Hybrid decoder stack (×8 blocks)
Reader B stream (h_B, Δ_B)
Sibling A
K_A, V_A
skip GDN
0-copy
GDN 4k+1
h_B → h'_B
GDN 4k+2
skip Span A
GDN 4k+3
private state
24
GDN
Full-Attn 4k+4
K_0
K_A
K_B
8
Attn
97.3% retrieval acc. · 2.7% hijack (21×↓ vs. text)
(c) Monotonic RoPE jump
1D token position axis (e_A > p_B)
0
p_fork
p_B
e_A
201+
Unshifted (Rule K1)
0.459 EM
c_B
q_B
q_B < k_A
Span A [s_A..e_A)
RoPE re-rotation (Rule K3)
O(N²) copy
c_B
R_Δθ Copy(K_A)
Δ_B
Octopus monotonic jump (K2)
0.630 EM
jump
c_B
Δ_B @ 201+
0-copy K_A (native pos)
s(c'_B) = max(p_B, e_A + 1) · causal q_B > k